

Circles

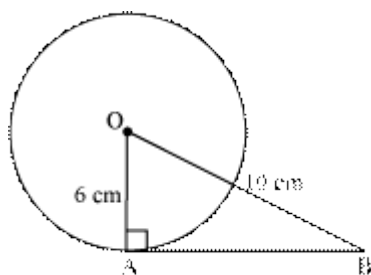
- **Concept of tangent at any point of the circle**

Theorem: The tangent at any point on a circle is perpendicular to the radius through the point of contact.

Example:

A tangent AB at a point A of a circle of radius 6 cm meets a line through the centre O at the point B, such that OB = 10 cm. Find the length of AB.

Solution:



It is known that the tangent at any point on a circle is perpendicular to the radius through the point of contact.

$OA \perp AB$

By applying Pythagoras theorem in right triangle OAB, we obtain

$$OA^2 + AB^2 = OB^2$$

$$\Rightarrow 6^2 + AB^2 = 10^2$$

$$\Rightarrow AB^2 = (100 - 36) \text{ cm}^2$$

$$\Rightarrow AB^2 = 64 \text{ cm}^2$$

$$\Rightarrow AB = \sqrt{64} \text{ cm} = 8 \text{ cm}$$

No tangent can be drawn to a circle passing through a point lying inside the circle.

One and only one tangent can be drawn to a circle passing through a point lying on the circle.

Exactly two tangents can be drawn to a circle through a point lying outside the circle.

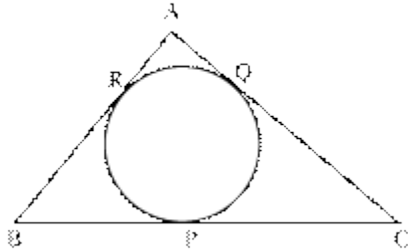
- **Tangent drawn from an external point to a circle**

Length of the tangent: The length of the segment of the tangent from an external point P to the point of contact with the circle is called the length of the tangent from the point P to the circle.

Theorem: The lengths of tangents drawn from an external point to a circle are equal.

Example:

In the given figure, a circle is inscribed in $\triangle ABC$ touching the points, P, Q, and R.



If $AB = 7$ cm, $BC = 9$ cm, $CA = 8$ cm, then find the measures of AR, AQ, BR, BP, CP, and CQ.

Solution:

It is known that the lengths of tangents drawn from an external point to a circle are equal.

$$AR = AQ = a \text{ (say)}$$

$$BR = BP = b \text{ (say)}$$

$$CP = CQ = c \text{ (say)}$$

$$AB + BC + CA = (7 + 9 + 8) \text{ cm} = 24 \text{ cm}$$

$$\Rightarrow (a + b) + (b + c) + (c + a) = 24 \text{ cm}$$

$$\Rightarrow 2(a + b + c) = 24 \text{ cm}$$

$$\Rightarrow a + b + c = 12 \text{ cm}$$

$$AB = 7 \text{ cm}$$

$$\Rightarrow a + b = 7 \text{ cm}$$

$$\therefore c + 7 \text{ cm} = 12 \text{ cm}$$

$$\Rightarrow c = (12 - 7) \text{ cm} = 5 \text{ cm}$$

$$BC = 9 \text{ cm}$$

$$\Rightarrow b + c = 9 \text{ cm}$$

$$\Rightarrow b = 9 - c = (9 - 5) \text{ cm} = 4 \text{ cm}$$

$$a + b + c = 12 \text{ cm}$$

$$\therefore 9 \text{ cm} + a = 12 \text{ cm}$$

$$\Rightarrow a = (12 - 9) \text{ cm} = 3 \text{ cm}$$

Hence, $AR = AQ = 3$ cm,

$BR = BP = 4$ cm,

$$CP = CQ = 5 \text{ cm.}$$

Results: If two tangents are drawn to a circle from an external point, then

- 1.
- 1.
2. 1. they subtend equal angles at the centre.
3. 2. they are equally inclined to the segment, joining the centre to that point.

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